

6 Differentiation

What students need to learn:

Applications of differentiation to maxima and minima and stationary points, increasing and decreasing functions.

The notation $f''(x)$ may be used for the second order derivative.

To include applications to curve sketching. Maxima and minima problems may be set in the context of a practical problem.

7 Integration

What students need to learn:

Evaluation of definite integrals.

Interpretation of the definite integral as the area under a curve.

Students will be expected to be able to evaluate the area of a region bounded by a curve and given straight lines.

Eg find the finite area bounded by the curve $y = 6x - x^2$ and the line $y = 2x$.

$\int x \, dy$ will not be required.

Approximation of area under a curve using the trapezium rule.

For example,

evaluate $\int_0^1 \sqrt{2x+1} \, dx$

using the values of $\sqrt{2x+1}$ at $x = 0, 0.25, 0.5, 0.75$ and 1 .

Unit C3

Core Mathematics 3

A2 compulsory unit for GCE Mathematics and GCE Pure Mathematics
Mathematics

C3.1 Unit description

Algebra and functions; trigonometry; exponentials and logarithms; differentiation; numerical methods.

C3.2 Assessment information

Prerequisites and preamble

Prerequisites

A knowledge of the specifications for C1 and C2, their preambles, prerequisites and associated formulae, is assumed and may be tested.

Preamble

Methods of proof, including proof by contradiction and disproof by counter-example, are required. At least one question on the paper will require the use of proof.

Examination

The examination will consist of one 1½ hour paper. It will contain about seven questions of varying length. The mark allocations per question which will be stated on the paper. All questions should be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: +, −, ×, ÷, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Formulae which students are expected to know are given below and these will not appear in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

This section lists formulae that students are expected to remember and that may not be included in formulae booklets.

Trigonometry

$$\cos^2 A + \sin^2 A \equiv 1$$

$$\sec^2 A \equiv 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A \equiv 1 + \cot^2 A$$

$$\sin 2A \equiv 2 \sin A \cos A$$

$$\cos 2A \equiv \cos^2 A - \sin^2 A$$

$$\tan 2A \equiv \frac{2 \tan A}{1 - \tan^2 A}$$

Differentiation

function

$$\sin kx$$

$$\cos kx$$

$$e^{kx}$$

$$\ln x$$

$$f(x) + g(x)$$

$$f(x) g(x)$$

$$f(g(x))$$

derivative

$$k \cos kx$$

$$-k \sin kx$$

$$k e^{kx}$$

$$\frac{1}{x}$$

$$f'(x) + g'(x)$$

$$f'(x) g(x) + f(x) g'(x)$$

$$f'(g(x)) g'(x)$$

1 Algebra and functions

What students need to learn:

Simplification of rational expressions including factorising and cancelling, and algebraic division.

Denominators of rational expressions will be linear or quadratic, eg $\frac{1}{ax+b}$,

$$\frac{ax+b}{px^2+qx+r}, \frac{x^3+1}{x^2-1}.$$

Definition of a function. Domain and range of functions. Composition of functions. Inverse functions and their graphs.

The concept of a function as a one-one or many-one mapping from $\mathbb{R}$ (or a subset of $\mathbb{R}$) to $\mathbb{R}$. The notation $f: x \mapsto$ and $f(x)$ will be used.

Students should know that fg will mean 'do g first, then f '.

Students should know that if f^{-1} exists, then $f^{-1}f(x) = ff^{-1}(x) = x$.

The modulus function.

Students should be able to sketch the graphs of $y = |ax+b|$ and the graphs of $y = |f(x)|$ and $y = f(|x|)$, given the graph of $y = f(x)$.

Combinations of the transformations $y = f(x)$ as represented by $y = af(x)$, $y = f(x) + a$, $y = f(x+a)$, $y = f(ax)$.

Students should be able to sketch the graph of, for example, $y = 2f(3x)$, $y = f(-x) + 1$, given the graph of $y = f(x)$ or the graph of, for example, $y = 3 + \sin 2x$,

$$y = -\cos\left(x + \frac{\pi}{4}\right).$$

The graph of $y = f(ax+b)$ will *not* be required.